

Chapter 10 — Use cases in healthcare and finance

Synthesis methods become concrete in domain settings

Learning objectives

- By the end of this part

Healthcare privacy barriers

- Healthcare data face strict privacy rules in the United States and Europe
- Real electronic health records often cannot be shared freely for external research or
- Provided re-identification risk is validated, not assumed away

Example 10.12 — Privacy-Preserving Medical Records

- Example 10.12 — hands-on module
- Visit counts
- External researchers prototype models without accessing raw patient files
- Explore the chapter example module
- View files: ``modules/chapter10/example12/``

Simulating rare conditions

- Low-prevalence diseases yield too few real cases for robust machine learning
- Synthetic generation can augment rare oncology or genetic presentations while preserving
- The goal is better classifiers and simulators

Example 10.13 — Rare-Disease Case Simulation

- Example 10.13 — hands-on module
- Example 10.13 shows an oncology team generating additional synthetic cases of a rare
- Explore the chapter example module
- View files: ``modules/chapter10/example13/``

Finance and confidentiality

- Financial institutions handle sensitive transactional data restricted by security and
- Synthetic data offers an alternative for fraud modeling and strategy simulation without

Fraud detection with synthetic patterns

- Fraud is a rare-event problem: historical logs contain few positive examples
- Synthetic generators can emit diverse fraudulent sequences

Example 10.14 — Synthetic Fraud Transaction Patterns

- Example 10.14 — hands-on module
- Example 10.14 describes a payments provider synthesizing card-not-present fraud sequences
- Detectors gain positive examples absent from historical logs
- Explore the chapter example module
- View files: ``modules/chapter10/example14/``

Trading simulations

- Edge conditions underrepresented in limited historical windows
- Synthetic finance use cases still require validation against live macro regimes before

Takeaways

- Healthcare synthesis targets privacy-preserving sharing and rare-disease coverage
- Both domains demand regulatory awareness and fidelity checks beyond aggregate similarity
- Autonomy and cybersecurity use cases follow in the next part

Next

- Complete the quiz for this part
- Ransomware attack simulation for cybersecurity training